Preface

The field of geospatial analysis is undergoing a rapid transformation, driven by advancements in digital mapping, web-based platforms, and artificial intelligence. This preface sets the stage for understanding the profound impact of these technologies on geospatial analysis and how they are making complex spatial data more accessible and actionable for a broader audience.

The motivation for this study stems from the need to simplify the interaction with geospatial data, making it more intuitive and accessible to non-experts. Traditional geospatial analysis tools often require specialized knowledge and expertise, creating a barrier for many potential users. The integration of AI technologies, particularly natural language processing (NLP), presents an opportunity to democratize access to these powerful tools, enabling a wider range of users to leverage spatial data for various applications, from urban planning to environmental monitoring.

The development of the GeoInsight application serves as a case study in this transformation. GeoInsight combines the power of NLP, advanced data visualization, and seamless integration with GIS software to provide a user-friendly platform for geospatial analysis. By allowing users to interact with geospatial data using natural language queries, GeoInsight removes the complexity of traditional query languages, making geospatial analysis accessible to everyone.

This study explores the various aspects of GeoInsight, from its development using Next.js, Node.js, and Tailwind CSS to its integration with the Overpass API for data retrieval from OpenStreetMap. It highlights the application's ability to translate natural language inputs into structured queries, visualize data through customizable layers, and export data in GeoJSON format for further analysis in advanced GIS software like ArcGIS and QGIS.

One of the key motivations behind this study is to demonstrate the potential of AI-driven geospatial tools to transform how spatial data is utilized. By providing intuitive and interactive interfaces, these tools can empower users from various fields to make informed decisions based on comprehensive spatial analysis. For example, urban planners can use GeoInsight to assess the distribution of green spaces and public amenities, while environmental scientists can analyze the impact of urban development on natural resources.